

Session Recap: Golden Triangle & Plan Revision

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Abstract

- HTML `
` tags in markdown weren't rendering properly in PDFs

1. Session Recap: Golden Triangle & Plan Revision

1.1 What We Accomplished

1. Identified the Problem

- HTML `<div style="...">` tags in markdown weren't rendering properly in PDFs
- User recognized this was causing "enough problems" and needed a proper solution
- Conversion chain: Markdown (with HTML) → HTML → PDF was breaking

2. Built the Golden Triangle System

Created: `src/waft/evolution/golden_triangle.py`

A unified conversion system providing clean, bidirectional conversion between:

- **Markdown ↔ HTML ↔ PDF**

Features:

- Markdown → HTML (preserves HTML blocks via `md_in_html` extension)
- HTML → PDF (clean WeasyPrint conversion)
- HTML → Markdown (round-trip capability)
- Markdown → PDF (direct path via HTML intermediate)
- Handles inline styles gracefully
- Style presets: premium, clinical_standard, professional

1.2 Key Insights

1. **Root Cause Solutions:** Instead of patching HTML div issues, we built a proper conversion system. Better engineering.
2. **Foundation First:** Building foundational systems (golden triangle) before implementing features (D&D campaign) is the right approach.

Integration:

- `PDFGenerator` now supports `use_golden_triangle=True` for direct markdown→PDF
- Backward compatible with existing ChatDistiller/TwoPageGenerator path
- Tested successfully with DnD_Preflight document

3. Tested the System

- Generated PDFs using golden triangle:
- `DnD_Preflight_Golden_Triangle.pdf` (direct conversion)
- `DnD_Preflight_FINAL.pdf` (via `PDFGenerator`)
- HTML divs now render correctly in PDFs
- Conversion chain works cleanly

4. Reflection & Documentation

- Wrote reflection entries in journal about:
- The HTML-in-markdown problem
- Building the golden triangle solution
- Preparing for plan revision
- Documented the system in code and journal

3. **Plan Evolution:** Plans need revision as capabilities evolve. The D&D campaign plan was created before golden triangle existed.

4. **Systematic Approach:** Problem → Solution → Test → Document → Revise Plan

1.3 Next Steps

1. **Revise D&D Campaign Plan:** Update to incorporate golden triangle for PDF generation
2. **Document Golden Triangle:** Add to project documentation

1.4 Files Created/Modified

Created:

- `src/waft/evolution/golden_triangle.py` - Golden triangle conversion system
- `_temp_pdf_samples/dnd_preflight_world_models.html` - HTML output for inspection
- `_temp_pdf_samples/session_recap_2026-01-12.md` - This recap

3. **Ready for Implementation:** D&D campaign can now use golden triangle for all PDFs
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Modified:

- `src/waft/evolution/pdf_generator.py` - Added golden triangle integration
- `_pyrite/journal/ai-journal.md` - Reflection entries

PDFs Generated:

- `~/Desktop/DnD_Preflight_Golden_Triangle.pdf`
- `~/Desktop/DnD_Preflight_PDFGenerator_GT.pdf`
- `~/Desktop/DnD_Preflight_FINAL.pdf`

1.5 Technical Details

Golden Triangle API

```
from src.waft.evolution.golden_triangle import GoldenTriangle

converter = GoldenTriangle()

# Markdown → HTML
html = converter.markdown_to_html(markdown_text, preserve_html=True)

# HTML → PDF
pdf_path = converter.html_to_pdf(html_content, output_path, style='premium')

# Markdown → PDF (direct)
pdf_path = converter.markdown_to_pdf(markdown_text, output_path, style='premium')

# HTML → Markdown (round-trip)
markdown = converter.html_to_markdown(html_content)
```

PDFGenerator Integration

```
from src.waft.evolution.pdf_generator import PDFGenerator

# Use golden triangle for direct markdown→PDF
generator = PDFGenerator.from_content(
    content=markdown_content,
    title="My Document",
    style="premium",
    use_golden_triangle=True # New option
)

pdf_file = generator.save(output_path, open_pdf=True)
```
